CIVIL ENGINEERING LICENSURE EXAMINATIONS
HYDRAULICS and GEOTECHNICAL ENGINEERING
NOVEMBER 2025

1. CE Board Nov. 2025

A gate is inclined at 45° to the horizontal and submerged in water. The center of pressure lies ____.

a) below the centroid ☜
b) at the centroid
c) above the centroid
d) at the top edge

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2. CE Board Nov. 2025

If the depth below a liquid surface is doubled, the hydrostatic pressure will:

a) double ☜
b) stay the same
c) become zero
d) halve

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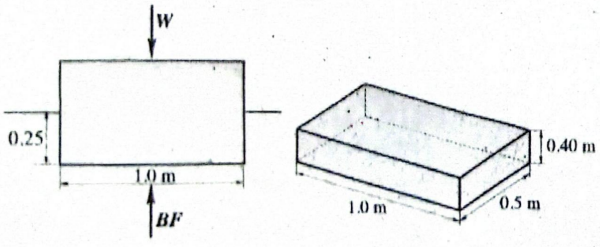
3. CE Board Nov. 2025

A rectangular wooden box 1 m x 0.5 m x 0.40 m floats in sea water with 0.25 m submerged. Specific gravity of sea water is 1.03. What is the weight of the box in N?



Solution:

BF = 1(0.5)(0.25)(1.03)(1000)(9.81)
BF = 128.75 kg
BF = 128.75(9.81) = ***1263 N***

4. CE Board Nov. 2025

A 50 m long pipe carries water at 2 m/sec. The pipe diameter is 100 mm and the friction factor f = 0.02. What is the head loss in meters?

Solution:

$$h = \frac{fLV^2}{2g\,D}$$

$$h = \frac{0.02(50)(2)^2}{2(9.81)(0.10)} = \mathbf{2.04\ m}$$

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5. CE Board Nov. 2025

Estimate the velocity head in meters of a liquid inside a pipe if its velocity is 5 m/sec.

Solution:

$$h = \frac{V^2}{2g}$$

$$h = \frac{(5)^2}{2(9.81)} = \mathbf{1.27\ m}$$

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6. CE Board Nov. 2025

A balloon filled with hot air has lower density than ambient air. Why does it rise?

a) Due to buoyant ☜
b) Because of surface tension
c) Because it has a higher pressure
d) Due to gravity pulling harder on cold air

7. CE Board Nov. 2025

A pump must raise water 12 m to a rooftop tank. What minimum pressure head should it provide in meters?

Solution:

To raise water 12 m to a rooftop, the minimum pressure head that the pump must provide is 12 m.

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8. CE Board Nov. 2025

What is the following quantities is conserved in ideal hydrodynamics flow?

a) Mass ☜ c) Pressure

b) Temperature d) Viscosity

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9. CE Board Nov. 2025

In a hydropower plant, water drops 80 m to spin a turbine. The plant operates with 85% efficiency and generates 500 KW of electric power. What is the minimum flow rate of water required in m^3/sec?

Solution:

$\rho = 1000\ kg/m^3$ (water density)

$P = 500\ KW$

$P = 500{,}000$ watts (power)

$$Q = \frac{\text{Power}}{\text{Eff. x Density x 9.81 x ht. of } H_2O \text{ drops}}$$

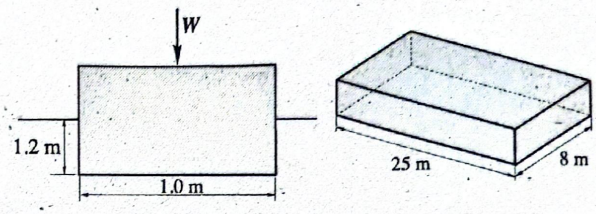
$$Q = \frac{500{,}000}{0.85(1000)(9.81)(80)}$$

$Q =$ ***0.75 m^3/sec.***

10. CE Board Nov. 2025

A barge measures 25 m long, 8 m wide, and sinks 1.2 m into sea water. Estimate the total weight of the barge including its cargo in kN.

Solution:



WF = BF

Density of sea water = $1025\ kg/m^3$

W = 25(8)(1.2)(1025)(9.81)

W = 2413260 N

W = ***2413 kN***

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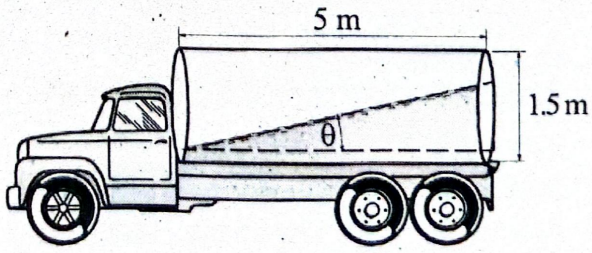
11. CE Board Nov. 2025

A closed cylindrical tank laid horizontally on a tank is 1.5 m in diameter and 5 m long. It is filled with water and the truck accelerates forward at $3\ m/sec^2$. What is the angle that the free surface of the water makers with the horizontal?

Solution:



$$\tan\theta = \frac{a}{g}$$

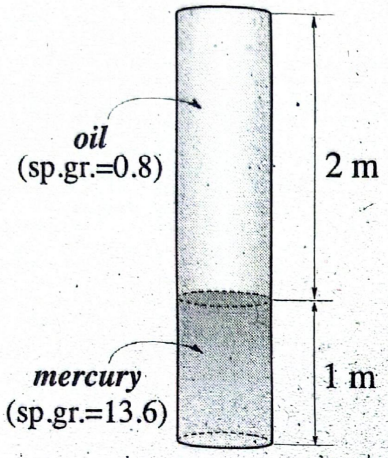
$$\tan\theta = \frac{3}{9.81}$$

$\theta =$ ***17°***

12. CE Board Nov. 2025

A composite fluid column has oil (sp.gr. = 0.8) for 2 m above the mercury for 1 meter. What is the total pressure at the bottom?

Solution:



P = wh

P = 0.8(9.81)(2) + 1(13.6)(9.81)

P = ***149.11 kPa***

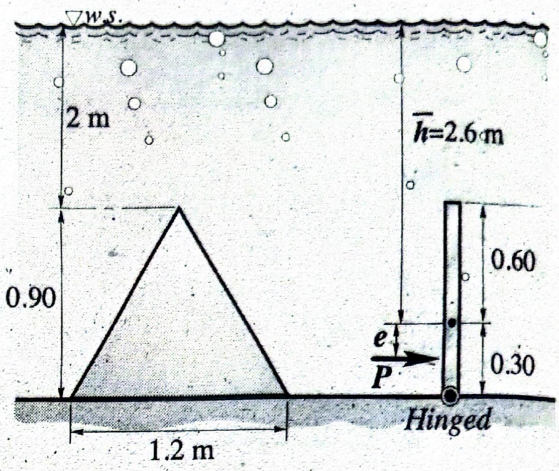
13. CE Board Nov. 2025

A triangular plate of height 0.9 m and base 1.2 m is installed vertically with fresh water on one side with its vertex at the top and the base is parallel to the water surface. The vertex is at a depth of 2 m below the water surface.

a) Evaluate the total hydrostatic force on the gate in kN.
b) Locate the point of action of the total hydrostatic force from the vertex in meters.
c) If the gate is hinged at the bottom, evaluate the force normal to the gate at its vertex that will be required to open it, in kN.

Solution:

a) Total hydrostatic force on the gate

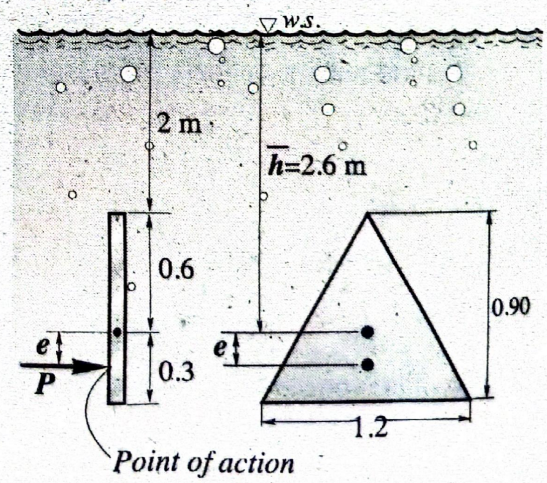


$$P = \gamma \bar{h} A$$

$$P = \frac{9.81(2.6)(0.9)(1.2)}{2}$$

P = ***13.77 kN***

b) Point of action of the total hydrostatic force from the vertex



$$I_g = \frac{bh^3}{36}$$

$$I_g = \frac{1.2(0.9)^3}{36}$$

I_g = 0.0243

$S_s = A\bar{h}$

$S_s = \frac{1.2(0.9)(2.6)}{2}$

$S_s = 1.404$

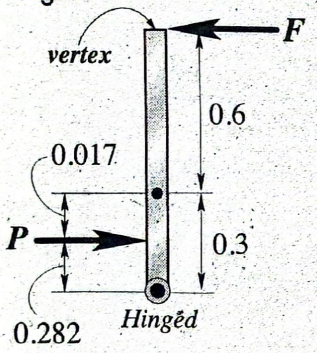
$e = \frac{I_g}{S_s}$

$e = \frac{0.0243}{1.404}$

e = 0.017

Point of action from the vertex
= 0.6 + 0.017
= ***0.612 m***

c) Force normal to the gate at its vertex to open the gate



∑M at the hinge
F(0.9) = P(0.283)
F(0.9) = 13.77(0.283)
F = ***4.33 kN***

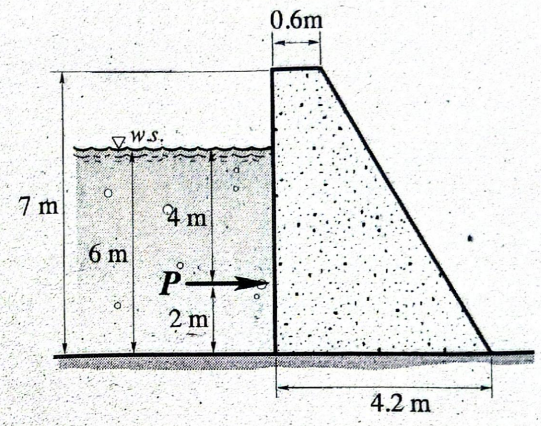
14. CE Board Nov. 2025

A concrete dam of trapezoidal cross section with one face vertical has a thickness of 0.60 m at the top and 4.2 m at the bottom. It is 7 m high and has a horizontal base. The vertical face is subjected to water pressure, the water standing 6 m above the base. The weight of the concrete is 24 kN/m³. The coefficient of friction between the concrete and the soil is 1.34 per meter length of the dam, estimate the following:

a) Total hydrostatic force on the dam in kN
b) The resulting moment of the dam to overturning in kN.m.
c) Factor of safety against sliding

Solution:

a) Total hydrostatic force on the dam



$P = \frac{1}{2} wh^2$

$P = \frac{1}{2} (9.81)(6)^2$

P = ***176.58 kN***

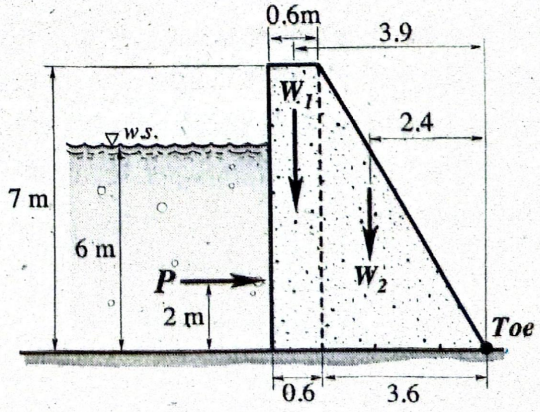
or

$P = w\,\bar{h}\,A$

P = 9.81(3)((6)(1)

P = 176.58

b) Resulting moment of the dam to overturning



Considering 1 meter length of the dam.

$W_1 = 0.60(1)(7)(24)$

$W_1 = 100.8$ kN

$$W_2 = \frac{7(3.6)(1)(24)}{2}$$

$W_2 = 302.40$ kN

Resulting moment of the dam

$M_1 = W_1 (3.9) + W_2 (2.4)$

$M_1 = 100.8(3.9) + 302.40(2.4)$

$M_1 = 1118.88$ say ***1118.90***

c) Factor of safety against sliding

Total weight = $W_1 + W_2$

Total weight = 100.8 + 302.40

Total weight = 403.20 kN

Resulting force = 1.34(403.20)

Resulting force = 540.29 kN

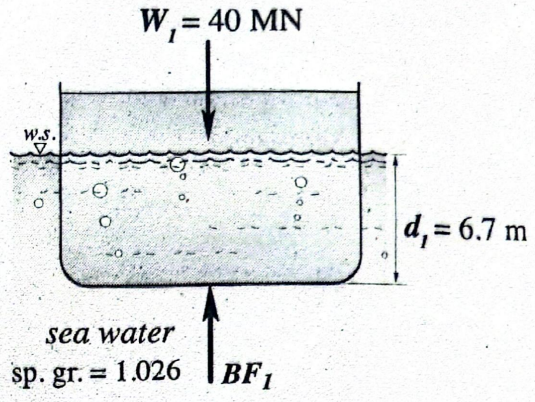
Factor of safety against sliding

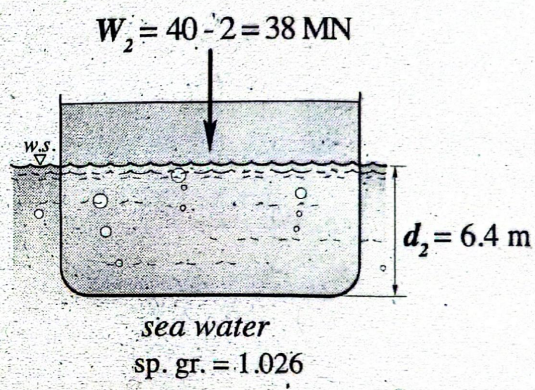
$$= \frac{540.29}{176.58} = \mathbf{3.06}$$

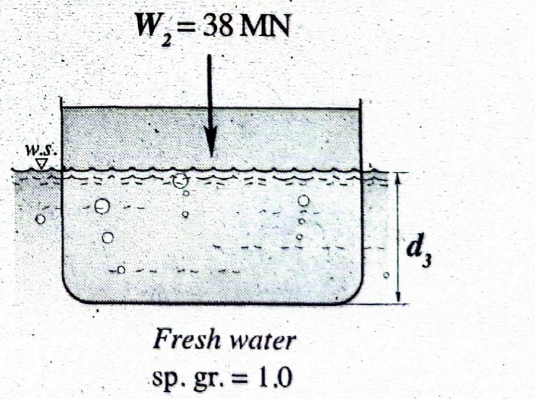
15. CE Board Nov. 2025

A ship with vertical sides near the water line weighs 40 MN including its cargo and has a draft of 6.7 m in seawater (sp.gr. = 1.026). Unloading 2 MN of each cargo, the draft decreases to 6.4 m with its cargo reduced, the ship enters a harbor of fresh water. Evaluate the draft of the ship in fresh water in meters.

Solution:







At seawater, sp.gr. = 1.026 $d_2 = 6.4$ m

$W_2 = \text{sp.gr.}(9.81)(A)d_2$

$$A = \frac{W_2}{sp.gr.(9.81)\,d_2}$$

$$A = \frac{W_2}{1.026(9.81)\,d_2}$$

$$A = \frac{W_2}{10.065\,d_2}$$

At fresh water (sp.gr. = 1.0) $d_3 = ?$

$W_2 = 1(9.81)A\, d_3$

$$d_3 = \frac{W_2}{9.81\, A}$$

$$d_3 = \frac{W_2}{\frac{9.81\, W_2}{10.065\, d_2}}$$

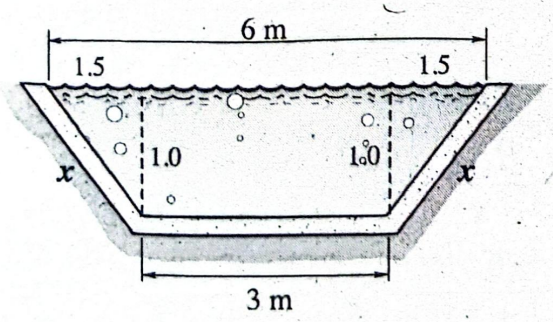
$$d_3 = \frac{10.065\, d_2}{9.81}$$

$$d_3 = \frac{10.065(6.4)}{9.81} = \mathbf{6.566\ m}$$

16. CE Board Nov. 2025

A 100 m long canal delivers 10 m^3/sec. of water to irrigate a rice field. The slope is 0.0008, and Manning n = 0.018. What is the required hydraulic radius R (in meters) to maintain this discharge in a trapezoidal channel with side slopes 1.5H:1V and bottom width of 3 m?

Solution:



$$A = \left(\frac{6+3}{2}\right)(1)$$

$A = 4.5\ m^2$

$Q = AV$

$10 = 4.5\ V$

$V = 2.22$ m/s

$$V = \frac{1}{n}\ R^{2/3}\, S^{1/2}$$

$$2.22 = \frac{1}{0.018}\ R^{2/3}(0.0008)^{1/2}$$

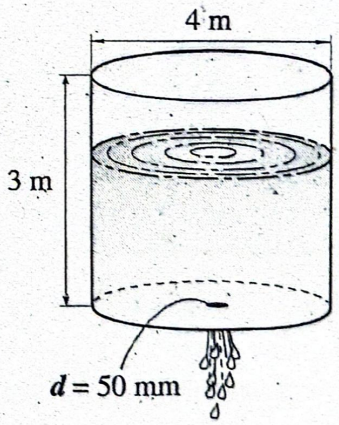
$$R^{2/3} = 1.413$$

R = ***1.68***

17. CE Board Nov. 2025

A right cylindrical container 4 m in diameter is 3 m tall. A 50 mm diameter note is provided at the bottom of the container to drain the water when necessary. If it is full of water, how long in hours, does it take to empty the container from the instant the hole is opened? Coeff. of velocity and coeff. of contraction both equal to 1.0.

Solution:



$$A_s = \frac{\pi}{4}(4)^2$$

$A_s = 4\pi$

$$A = \frac{\pi}{4}(0.05)^2$$

$A = 0.0196$

$$t = \frac{2\, A_s \sqrt{h}}{CA\sqrt{2g}}$$

$$t = \frac{2(4)\pi\sqrt{3}}{1.0(0.00196)\sqrt{2(9.81)}}$$

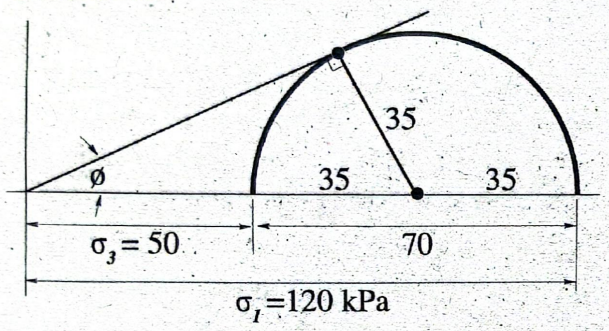
$t = 5014$ sec.

$$t = \frac{5014}{3600} = \mathbf{1.39\ hours}$$

18. CE Board Nov. 2025

A specimen of soil under triaxial test was consolidated under a chamber confining pressure of 50 kPa. The axial stress on the specimen was then increased and failure occurred when the axial stress reached 120 kPa. Estimate the angle of internal friction of the clay in degrees

Solution:



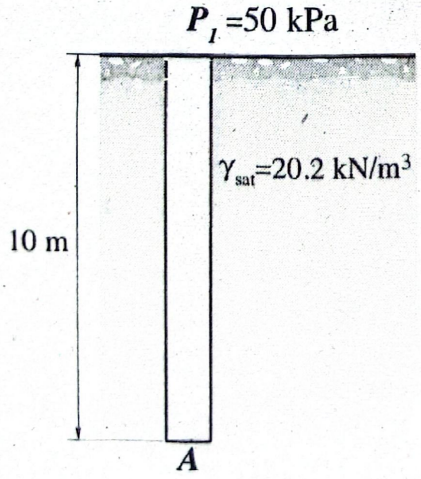
$$\sin ø = \frac{35}{50 + 35}$$

ø = ***24.3°***

19. CE Board Nov. 2025

A borehole in Lucena City reveals a 10 m thick saturated soi profile. If surcharge stress from a nearby warehouse is 50 kPa, what is the effective stress at 10 m depth. γ_{sat} = 20.2 kN/m^3

Solution:



Stress at A = 50 + (20.2 – 9.81)(10)
Stress at A = ***153.9 kPa***

20. CE Board Nov. 2025

A soil has a water content = 25%, void ratio = 0.6 and G_s = 2.7. What is the degree of saturation in percent?

Solution:

$$S = \frac{wG_s}{e}$$

$$S = \frac{25(2.7)}{0.6}$$

S = ***112.5%***

21. CE Board Nov. 2025

Evaluate the plastic settlement in meters, of a layer of plastic clay due to an increase of pressure caused by loads above it under the following conditions:
Initial intergranular pressure = 180 kPa
Increase in intergranular pressure = 140 kPa
Thickness of clay layer = 8 m
Coefficient of consolidation = 0.315
Void ratio of the soil = 1.132

Solution:

$$S = \frac{H}{1+e} C_c \log\left(\frac{P_o + \Delta P}{P_o}\right)$$

$$S = \frac{8(0.315)}{1 + 1.132} \log\left(\frac{180 + 140}{180}\right)$$

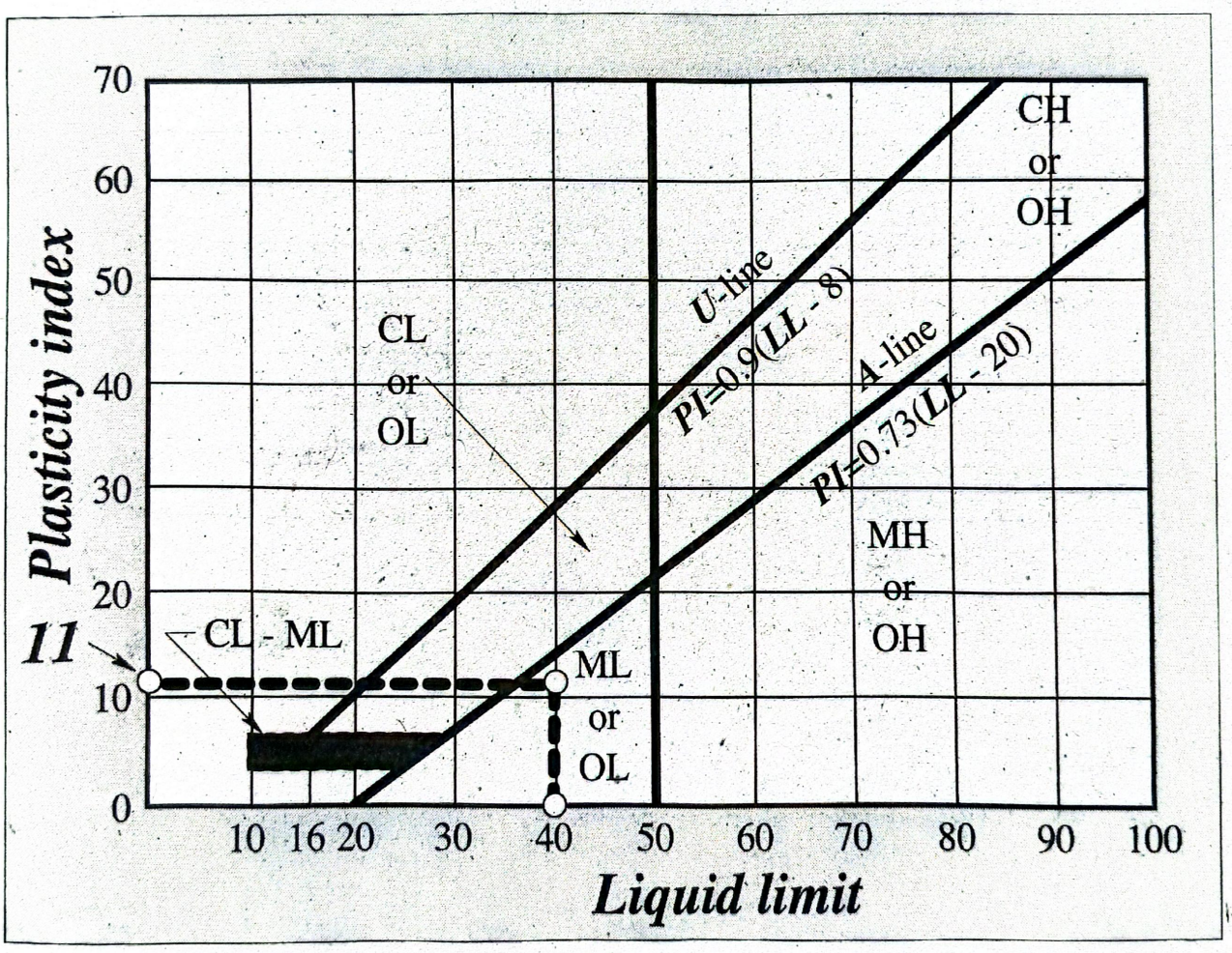
S = ***0.295 m***

Civil Engineering Licensure Examinations

22. CE Board Nov. 2025

Using the Casagrande Plasticity Chart, determine the classification of a soil with LL = 40% and PI = 11%.

Solution:

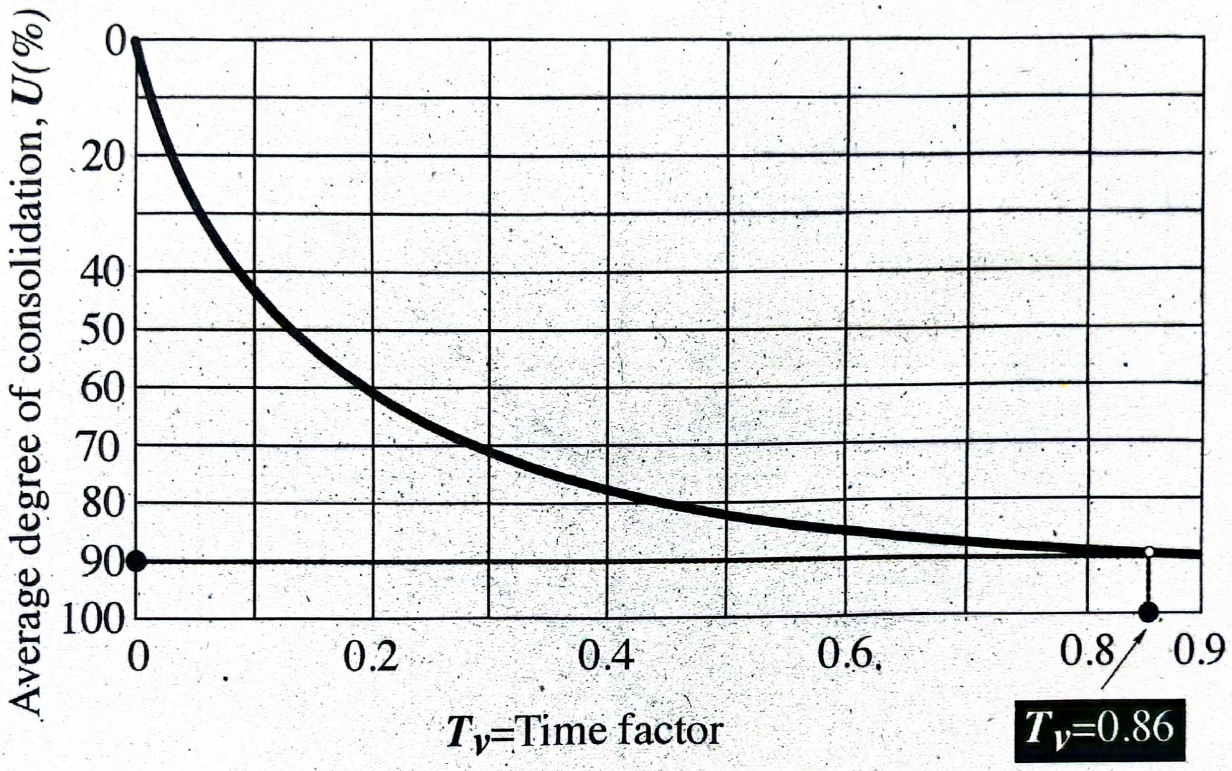


The soil is ML.

23. CE Board Nov. 2025

A 3 m thick layer (double-drainage layer) 90% consolidation occurred in 365 days. Estimate the coefficient of consolidation C_v in cm²/sec.

Solution:



$$t = \frac{T_v\,(H_{dr})^2}{C_v}$$

$$H_{dr} = \frac{3}{2} = 1.5\ m$$

H_{dr} = 150 cm

T_v = 0.86 from table

t = 365(24)(3600)

t = 31536000 sec.

$$t = \frac{T_v\,(H_{dr})^2}{C_v}$$

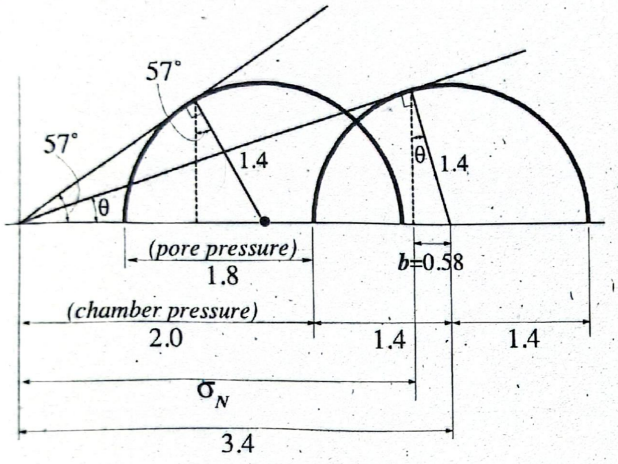
$$31536000 = \frac{0.86(150)^2}{C_v}$$

C_v = ***0.00061 cm²/sec.***

24. CE Board Nov. 2025

A consolidated undrained tri-axial test was performed at a saturated clay with a chamber pressure of 2 kPa. At failure, the difference between the normal pressure and confining pressure was 2.8 kPa. The pore water pressure was 1.8 kPa and the failure angle was 57 degrees. Calculate the normal stress on the failure surface in kPa.

Solution:



$$\sin\theta = \frac{1.4}{3.4}$$

$\theta = 24.32^\circ$

$$\sin 24.32^\circ = \frac{b}{1.4}$$

b = 0.58

Normal stress at the failúre surface:

$\sigma_N = 3.4 - 0.58$

σ_N = ***2.82 kPa***

25. CE Board Nov. 2025

A soil sample has a cross sectional area of 25 cm^2 and a length of 10 cm. During a falling head test, the head dropped from 50 cm to 25 cm in 60 seconds. If the stand pipe has a diameter of 1.0 cm, compute the coefficient of permeability in cm/sec.

Solution:

Coefficient of permeability of soil in cm/sec.:

$$K = \frac{aL}{At}\log_e\left(\frac{h_1}{h_2}\right)$$

L = 10 cm $\qquad$ h_1 = 50 cm

t = 60 sec. $\qquad$ h_2 = 25 cm

a = area of standpipe

$$a = \frac{\pi}{4}(1)^2 = 0.7854\ cm^2$$

A = 25 cm^2

$$K = \frac{aL}{At}\log_e\left(\frac{h_1}{h_2}\right)$$

$$K = \frac{0.7854(10)}{25(60)}\log_e\left(\frac{50}{25}\right)$$

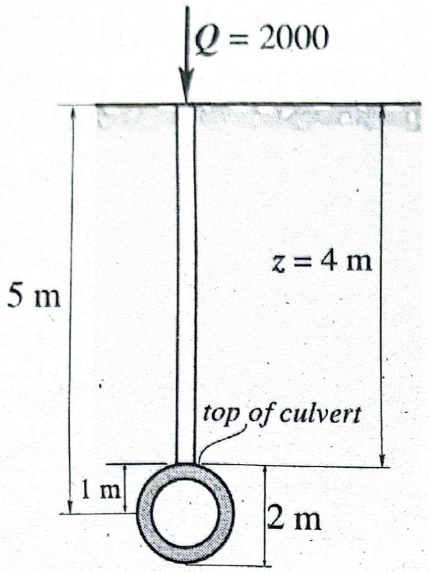
K = ***3.63 x 10^{-3} cm/sec.***

26. CE Board Nov. 2025

A pipe culvert of 2 m diameter is installed in a 5 m deep trench excavation of a subgrade of a propored roadway. Evaluate the pressure in kPa, on top of the culvert if the axial load of the road roller compacting the subgrade is 2000 kN on the assumption of a homogeneous, isotropic soil mass up to the level of the top of the culvert. Utilize Boussinesq's equation

$$P = \frac{0.477QN}{z^2}\ where\ N = \frac{1}{\left[1 + \left(\frac{r}{z}\right)^2\right]^{2.5}}$$

Solution:



$$N = \frac{1}{\left[1+\left(\frac{r}{z}\right)^2\right]^{2.5}}$$

$$N = \frac{1}{\left(1+\frac{0}{4}\right)^{2.5}}$$

N = 1

$$P = \frac{0.477QN}{z^2}$$

$$P = \frac{0.477(2000)(1)}{(4)^2}$$

P = ***59.6 kPa***

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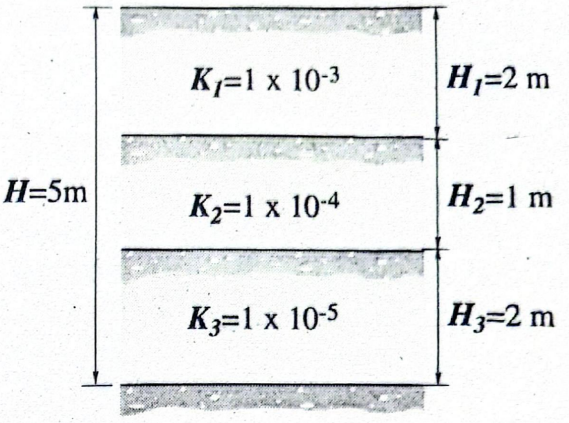
27. CE Board Nov. 2025

If a stratified soil profile has 3 horizontal layers with K_1 = 1 x 10^{-3} m/s (thickness 2 m), K_2 = 1 x 10^{-4} m/s (thickness is 1 m) and K_3 = 1 x 10^{-5} m/s (thickness = 2 m). What is the equivalent horizontal permeability in m/s?

Solution:



$$K_h = \frac{1}{H}(K_1\,h_1 + K_2\,H_2 + K_3\,H_3)$$

$$K_h = \frac{1}{5}[1\,x\,10^{-3}(2) + 1\,x\,10^{-4}(1) + 1\,x\,10^{-5}(2)]$$

K_h = ***4.24 x 10^{-4}***

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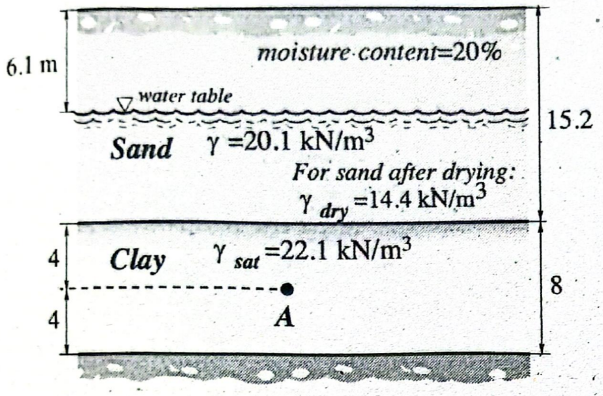
28. CE Board Nov. 2025

A clay layer 8 m thick is overlain with 15.2 m thick of sand (sp.gr. = 2.71). The water table is 6.1 m below the sand (ground) surface. The saturated unit weight of the clay is 22.1 kN/m^2. The sand below the water table has a unit weight of 20.1 kN/m^3. The sand above the water table has a moisture content of 20%. After drying, the sand was observed to have a dry unit weight of 14.4 kN/m^3.

a) Evaluate the unit weight of the moist sand above the water table in kN/m^3.
b) Determine the total stress at mid-height of the clay layer in kPa.
c) Determine the effective stress at mid-height of the clay layer in kPa.

Solution:

a) Unit weight of the moist sand above the water table



γ_{sand} = 20.1 kN/m³ below the water table

γ_{dry} of sand = 14.4 kN/m³ after drying

$$S = \frac{w\,G_s}{e} \quad \begin{pmatrix} degree\ of\ saturation \\ of\ sand \end{pmatrix}$$

$$\gamma_{dry} = \frac{G_s\,\gamma_w}{1+e}$$

$$14.4 = \frac{2.71(9.81)}{1+e}$$

e = 0.846 *void ratio of dry sand*

Degree of saturation of sand:

$$S = \frac{wG}{e} \qquad w = 20\%\ water\ content$$

$$S = \frac{20(2.71)}{0.846}$$

S = 64.07%

γ_{sat} of sand:

$$\gamma_{sat} = \frac{(G_s + e)\gamma_w}{1+e}$$

$$\gamma_{sat} = \frac{(2.71 + 0.846)(9.81)}{1 + 0.846}$$

γ_{sat} = 18.90 kN/m³

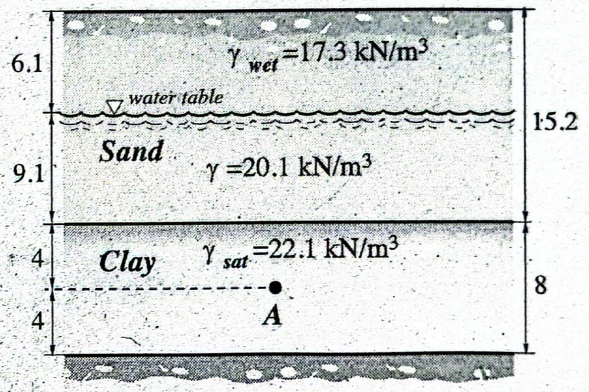
Moist unit weight of sand:

$$\gamma_{wet} = \frac{(G_s + Se)\gamma_w}{1+e}$$

$$\gamma_{wet} = \frac{[2.71 + 0.6407(0.846)]9.81}{1 + 0.846}$$

γ_{wet} = ***17.20 kN/m³***

b) Total stress at mid-height of the clay

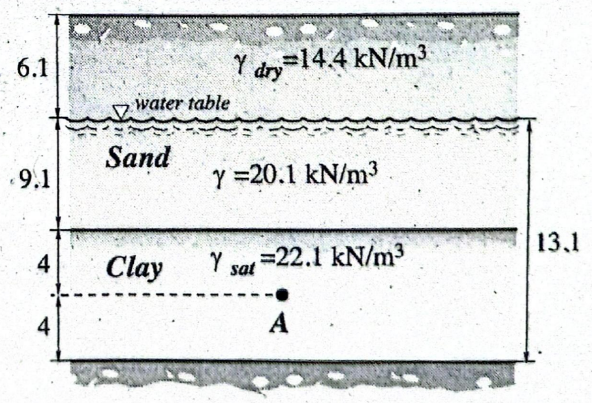


Total stress at A:

σ_A = 17.3(6.1) + 20.1(9.1) + 22.1(4)

σ_A = ***376.84 kPa***

c) Effective stress at mid-height of clay layer



Effective stress at A = 376.84 – 9.81(13.1)

Effective stress at A = ***248.3 kPa***

29. CE Board Nov. 2025

A soil has the following properties:

58% passing No. 200 sieve

LL = 60% PL = 25%

What is the classification under the Unified Soil Classification System (USCS)?

Solution:

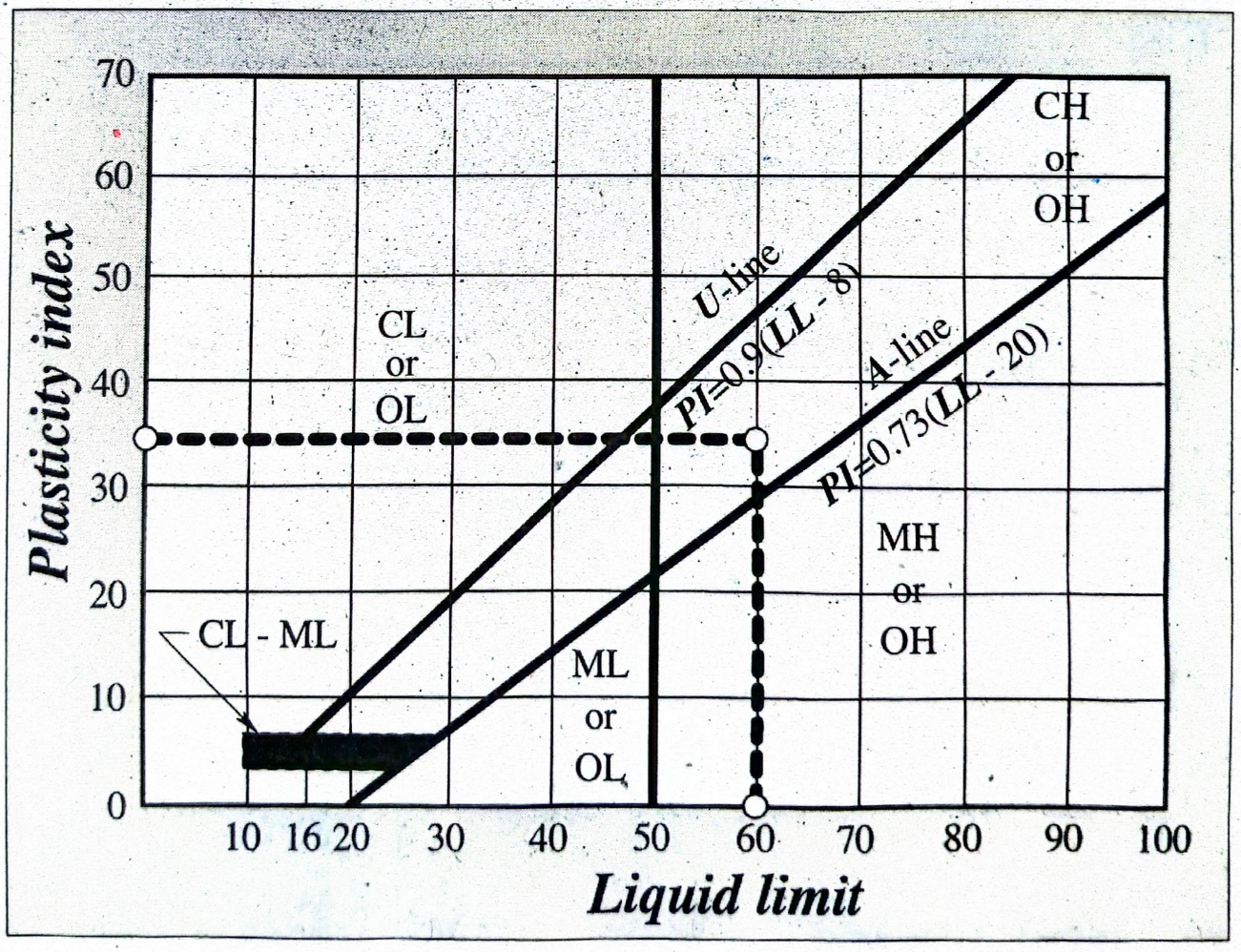
58% passing No. 200 sieve

LL = 60 PL = 25

Plasticity index P.I. = LL – PL

P.I. = 60 – 25 = 35

58% > 50%



Therefore, the soil is CH.

30. CE Board Nov. 2025

The maximum dry unit density up to which any soil can be compacted depends upon _____

a) both moisture content and amount of compaction energy ☜

b) amount of compaction energy only

c) moisture content only

d) none of the given choices

31. CE Board Nov. 2025

The neutral stress in a soil mass is ___

a) stress taken up by the pore water ☜

b) stress taken up by the soil particles

c) force per effective area

d) force per neutral area

32. CE Board Nov. 2025

Which are classifications of flow in fluid mechanics?

a) Gravity, Pumped, Pressurized

b) Transitional, Steady, Chaotic

c) Laminar, Turbulent, Uniform ☜

d) Steady, Unsteady, Natural

33. CE Board Nov. 2025

Among the methods for determining the shear strength of soils, which test may NOT accurately represent filed conditions?

a) Triaxial tests

b) Standard penetration test (SPT) ☜

c) Unconfined compressive tests

d) Direct shear tests

34. CE Board Nov. 2025

What laboratory test is commonly used to determine both cohesion and angle of internal friction?

a) Permeability test

b) Unconfined compression test

c) Compaction test

d) Direct shear test ☜

35. CE Board Nov. 2025

Among the methods for determining the shear strength of soils, which test measures the shear strength of a soil sample under various confining pressure?

a) Standard penetration test (SPT)

b) Unconfined compressive tests

c) Triaxial tests ☜

d) Direct shear tests

36. CE Board Nov. 2025

Which property of soil increases with an increase in compaction effort?

a) Void ratio

b) Friction angle ☜

c) Cohesion

d) Permeability

37. CE Board Nov. 2025

The specific weight of water is 1000 kg/m^3. To use this value the water must be

a) at 4°C temperature ☜

b) at all of the given conditions

c) at mean sea level

d) at normal pressure of 750 mm

38. CE Board Nov. 2025

Which test is commonly used in the Philippines to determine the in-situ density of compacted soil in roadworks?

a) Sieve Analysis
b) Triaxial Test
c) Direct Shear Test
d) Sand Cone Method ☜

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39. CE Board Nov. 2025

Which of the following best describes the reason why saturated clays fail at lower shear strength under rapid loading?

a) Pore water cannot drain ☜
b) Soil particles become denser
c) Effective stress increases
d) Pore pressure dissipates

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40. CE Board Nov. 2025

What type of soil typically has the highest angle of internal friction?

a) Peat
b) Silt
c) Clay
d) Gravel ☜

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41. CE Board Nov. 2025

A DPWH project requires identifying a soil type for slope stability analysis. The test results show $C_u = 2.8$ and $C_c = 4.2$. How will you describe this soil gradation? See Unified Soil Classification System

a) Poorly graded ☜
b) Uniformly graded
c) Gap-graded
d) Well-graded

Solution:

Using USCS Criteria for gradation

For sand, it is well-graded if:

$C_u \geq 6$ and $1 \leq C_c \leq 3$

For gravel, it is well-graded if:

$C_u \geq 4$ and $1 \leq C_c \leq 3$

when $C_u = 2.8$ and $C_c = 4.2$ then the soil does not meet the criteria for well-graded soil. Therefore, the soil is ***poorly graded***.

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42. CE Board Nov. 2025

A square footing 2 m x 2 m is founded at 1.2 m depth in a soil with $\gamma = 17.5$ kN/m^3, $\phi = 35^\circ$, $N_q = 41.4$, $N_\gamma = 42.4$. Compute q_{ult} using Terzaghi's Formula in kPa.

Solution:

Ultimate bearing pressure:

$q_{ult} = 1.3C N_c + q N_q + 0.40 \gamma B N_\gamma$

$q_{ult} = 0 + 1.2(17.5)(41.4) + 0.40(17.5)(2)(42.4)$

$q_{ult} =$ ***1463 kPa***

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43. CE Board Nov. 2025

In accordance with the Unified Soil Classification System, fined grained soils are considered highly plastic if the liquid limit is greater than what percentage?

Solution:

when:

LL < 50% the soil has low plasticity
LL > 50% the soil has high plasticity

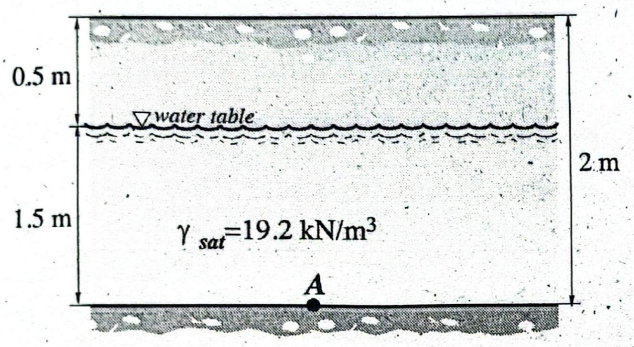
Therefore, the liquid limit is greater than 50% to be considered high plasticity.

44. CE Board Nov. 2025

A dam embankment in Laguna applies 80 kPa of stress on a saturated soil. The water table fluctuates seasonally between 0.5 m and 2.5 m. Estimate the range of effective stress at 2 m depth. γ_{sat} = 19.2 MPa

Solution:

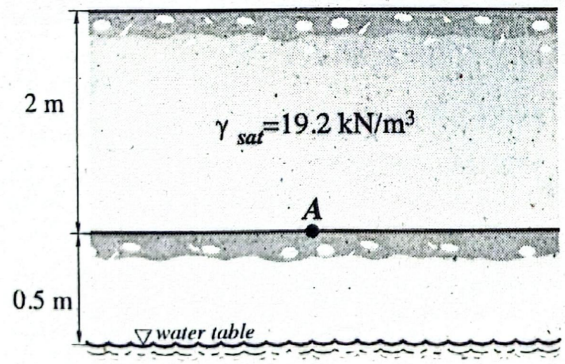
When water table is 0.5 depth



$\sigma_A = 19.2(2) - 9.81(1.5)$

σ_A = 23.69 kPa

When the water table is at 2.5 m depth



$\sigma_A = 19.2(2) = 38.4$ kPa

Range of effective stress is 23.69 kPa to 38.4 kPa.